

EBN111 SEMESTER TOETS/TEST 2 (14 MAY 2014)**ANTWOORDBLAD / ANSWER SHEET**

Van / Surname:	MEMO	Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language	Afrikaans			English			

Q1		Q2		Tot:	/45
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VRAAG 1 – QUESTION 1 (25)

1.1	$V_{X(3A)} = 6\text{ V}$		
1.2	$I_Y = 2\text{ mA}$		
1.3	$V_{TH} = 30\text{ V}$	$R_{TH} = 13\ \Omega$	
1.4	$I_N = -5\text{ A}$	$R_N = 0.8\ \Omega$	
1.5	$a) R_L = 10\ \Omega$	$b) P_{max} = 2\text{ W}$	
1.6	$a) V_X = -10\text{ V}$	$b) V_X = 10\text{ V}$	$c) V_X = 0\text{ V}$
1.7	$V_Y = 3.6\text{ V}$	$I_Y = 40\ \mu\text{A}$	
1.8	$V_Z = 14\text{ V}$	$I_Z = 2\text{ mA}$	
1.9	$V_I = -2\text{ V}$	$V_2 = 6.3\text{ V}$	$I_I = 25\ \mu\text{A}$

VRAAG 2 – QUESTION 2 (20)

2.1	$C_{eq} = 1.6\text{ F}$		
2.2	$a) v_C = 20\text{ mV}$	$b) w_L = 40\ \mu\text{J}$	
2.3	$a) i(t) = 66.67t^2\text{ mA}$	$b) w = 22.22\text{ mJ}$	
2.4	$a) f = 80\text{ Hz}$	$b) \phi = -160^\circ$	$c) \bar{V} = 5\angle 55^\circ\text{ V}$
	$d) \text{___} V_I \text{___} \text{leads___} V_2 \text{___} \text{by___} 50^\circ \text{___}$	$e) i(t) = 3\cos(50t + 105^\circ)$	
2.5	$Z_I = 2\angle 30^\circ\ \Omega$ (Polar format)		
2.6	$Z_{in} = 4.5 + j2\ \Omega$ (Rectangular format)		
2.7	$\bar{V}_1 = 5\angle -90^\circ\text{ V}$ (Polar format)		
2.8	$\bar{I}_1 = 7.32\angle -75^\circ\text{ A}$ (Polar format)		